

Linear algebra (official MATH 257 notes)

1. Introduction to Linear Systems

Definition 1.1

A linear equation is a equation of the form

$$a_1x_1 + \cdots + a_nx_n = b$$

where $a_1, \dots, a_n, b$ are numbers and $x_1, \dots, x_n$ are variables.

Definition 1.2

A linear system is a collection of one or more linear equations involving the same set of variables, say, $x_1, x_2, \dots, x_n$.

A solution of a linear system is a list $(s_1, s_2, \dots, s_n)$ of numbers that makes each equation in the system true when the values $s_1, s_2, \dots, s_n$ are substituted for $x_1, x_2, \dots, x_n$, respectively.

Theorem 1.3

A linear system has either one unique solution, no solution or infinitely many solutions.

Definition 1.4

The solution set of a linear system is the set of all solutions of the linear system. Two linear systems are equivalent if they have the same solution set.

2. Matrices

Definition 2.1

An $m \times n$ matrix is a rectangular array of numbers with m rows and n columns.

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

a_{ij} is in the i th row and j th column of A

Definition 2.2

Given a linear system,

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{aligned}$$

we define its coefficient and augmented matrices to be

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \quad \text{and} \quad \left[\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{array} \right],$$

respectively

Definition 2.3

An elementary row operation is one of the following:

- (Replacement) Add a multiple of one row to another row: $R_i \rightarrow R_i + cR_j$, where $i \neq j$,
- (Interchange) Interchange two rows: $R_i \leftrightarrow R_j$,
- (Scaling) Multiply all entries in a row by a nonzero constant: $R_i \rightarrow cR_i$, where $c \neq 0$.

Remark

Every row operation is reversible.

Definition 2.4

Two matrices are said to be row equivalent, if one matrix can be transformed into the other matrix by a sequence of elementary row operations.

Theorem 2.5

If the augmented matrices of two linear systems are row equivalent, then the two systems have the same solution set.

3. Echelon Forms of Matrices

Definition 3.1

A matrix is in echelon form (or row echelon form) if

1. all nonzero rows (rows with at least one nonzero element) are above any rows of all zeros, and
2. the leading entry (the first nonzero number from the left) of a nonzero row is always strictly to the right of the leading entry of the row above it.

Definition 3.2

A matrix is in row reduced echelon form (or: reduced echelon form, or RREF) if it is in echelon form and the following are satisfied:

1. the leading entry in each nonzero row is 1, and
2. each leading entry is the only nonzero entry in its column.

Theorem 3.3 (Uniqueness of RREF)

Each matrix is row-equivalent to one and only one matrix in reduced echelon form.

Definition 3.4

We say a matrix B is the reduced echelon form (or: the RREF) of a matrix A if A and B are row-equivalent and B is in reduced echelon form.

Definition 3.5

A pivot position is the position of a leading entry in an echelon form of a matrix. A pivot column is a column that contains a pivot position.

Definition 3.6

A basic variable (or pivot variable) is a variable that corresponds to a pivot column in the coefficient matrix of a linear system. A free variable is a variable that is not a pivot variable.

4. Gaussian Elimination

Goal. Solve linear systems for the pivot variables in terms of the free variables (if any) in the equation.

Algorithm 4.1 (Gaussian Elimination)

Given a linear system,

1. write down the augmented matrix,
2. find the reduced echelon form of the matrix,
3. write down the equations corresponding to the reduced echelon form, and
4. express pivot variables in terms of free variables.

Theorem 4.2

A linear system is consistent if and only if an echelon form of the augmented matrix has no row of the form $[0 \ \dots \ 0 \ b]$, where b is nonzero. If a linear system is consistent, then the linear system has

- a unique solution (when there are no free variables), or
- infinitely many solutions (when there is at least one free variable).

5. Linear Combinations

Definition 5.1

Consider $m \times n$ -matrices

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}, \quad \text{and} \quad B = \begin{bmatrix} b_{11} & b_{12} & \dots & b_{1n} \\ b_{21} & b_{22} & \dots & b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ b_{m1} & b_{m2} & \dots & b_{mn} \end{bmatrix}$$

- We define their sum $A + B$ to be

$$A + B := \begin{bmatrix} a_{11} + b_{11} & a_{12} + b_{12} & \dots & a_{1n} + b_{1n} \\ a_{21} + b_{21} & a_{22} + b_{22} & \dots & a_{2n} + b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} + b_{m1} & a_{m2} + b_{m2} & \dots & a_{mn} + b_{mn} \end{bmatrix}$$

- If $c \in \mathbb{R}$ is a scalar, we define the scalar product cA to be

$$cA := \begin{bmatrix} ca_{11} & ca_{12} & \dots & ca_{1n} \\ ca_{21} & ca_{22} & \dots & ca_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ ca_{m1} & ca_{m2} & \dots & ca_{mn} \end{bmatrix}$$

Remark

Addition is only defined if A has the same number of columns and rows as B .

Definition 5.2

A column vector is an $m \times 1$ -matrix. A row vector is a $1 \times n$ -matrix.

Definition 5.3

If A is $m \times n$, the transpose of A is the $n \times m$ matrix, denoted by A^T , whose columns are formed from the corresponding rows of A . In terms of matrix elements: $(A^T)_{ij} = A_{ji}$.

Remark

The transpose of a column vector is a row vector and vice versa.

Definition 5.4

The linear combination of $m \times n$ -matrices $A_1, A_2, \dots, A_p$ with coefficients $c_1, c_2, \dots, c_p$ is defined as:

$$c_1 A_1 + c_2 A_2 + \dots + c_p A_p.$$

Definition 5.5

Given $m \times n$ matrices $A_1, \dots, A_p$, we define their Span, $\text{Span}(A_1, \dots, A_p)$, to be the set of all linear combinations of $A_1, \dots, A_p$. That is,

$$\text{Span}(A_1, \dots, A_p) := \{c_1 A_1 + c_2 A_2 + \dots + c_p A_p : c_1, \dots, c_p \text{ scalars}\}$$

Definition 5.6

We denote the set of all column vectors of length m by $\mathbb{R}^m$.

Remark

Solving linear systems is the same as finding linear combinations!

Theorem 5.7

Let $a_1, \dots, a_n, b \in \mathbb{R}^m$. A vector equation

$$x_1 a_1 + x_2 a_2 + \dots + x_n a_n = b$$

has the same solution set as the linear system whose augmented matrix is

$$[a_1 \ a_2 \ \dots \ a_n \ | \ b]$$

In particular, b can be generated by a linear combination of $a_1, a_2, \dots, a_n$ if and only if there is a solution to the linear system corresponding to the augmented matrix.

6. Matrix-vector multiplication

Definition 6.1

Let x be a vector in $\mathbb{R}^n$ and $A = [a_1 \ \dots \ a_n]$ an $m \times n$ -matrix. We define the product Ax by

$$Ax = x_1 a_1 + x_2 a_2 + \dots + x_n a_n.$$

Remark

- Ax is a linear combination of the columns of A using the entries in x as coefficients.
- Ax is only defined if the number of entries of x is equal to the number of columns of A .

Theorem 6.2

Let $A = [a_1 \ \dots \ a_n]$ be an $m \times n$ -matrix and $b \in \mathbb{R}^m$. Then the following are equivalent:

- $(x_1, x_2, \dots, x_n)$ is a solution of the vector equation $x_1 a_1 + x_2 a_2 + \dots + x_n a_n = b$
- $\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$ is a solution to the matrix equation $Ax = b$.
- $(x_1, x_2, \dots, x_n)$ is a solution of the system with augmented matrix $[A \mid b]$

Remark

We will write $Ax = b$ for the system of equations with augmented matrix $[A \mid b]$.

7. Matrix Multiplication

Definition 7.1

Let A be an $m \times n$ -matrix and let $B = [b_1 \ \dots \ b_p]$ be an $n \times p$ -matrix. We define

$$AB := [Ab_1 \ Ab_2 \ \dots \ Ab_p]$$

Theorem 7.2

Let A be an $m \times n$ matrix and B be an $n \times p$ matrix. Then for every $x \in \mathbb{R}^p$

$$A(Bx) = (AB)x.$$

Note. Other ways to compute AB .

Row-Column Rule

Let A be $m \times n$ and B be $n \times p$ such that

$$A = \begin{bmatrix} R_1 \\ \vdots \\ R_m \end{bmatrix}, \quad \text{and} \quad B = [C_1 \ \dots \ C_p].$$

Then

$$AB = \begin{bmatrix} R_1 C_1 & \dots & R_1 C_p \\ R_2 C_1 & \dots & R_2 C_p \\ \vdots & \ddots & \vdots \\ R_m C_1 & \dots & R_m C_p \end{bmatrix}$$

and $(AB)_{ij} = R_i C_j = a_{i1} b_{1j} + a_{i2} b_{2j} + \dots + a_{in} b_{nj}$.

Outer Product Rule

Let A be $m \times n$ and B be $n \times p$ such that

$$A = [C_1 \ \dots \ C_n], \quad \text{and} \quad B = \begin{bmatrix} R_1 \\ \vdots \\ R_n \end{bmatrix}.$$

Then

$$AB = C_1 R_1 + \dots + C_n R_n$$

8. Properties of Matrix Multiplication

Definition 8.1

The identity matrix I_n of size n is defined as

$$I_n = \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{bmatrix}.$$

Theorem 8.2

Let A be an $m \times n$ matrix and let B and C be matrices for which the indicated sums and products are defined.

1. $A(BC) = (AB)C$ (associative law of multiplication)
2. $A(B + C) = AB + AC$, $(B + C)A = BA + CA$ (distributive laws)
3. $r(AB) = (rA)B = A(rB)$ for every scalar r ,
4. $A(rB + sC) = rAB + sAC$ for every scalars r, s (linearity of matrix multiplication)
5. $I_m A = A = A I_n$ (identity for matrix multiplication)

Remark

Properties above are analogous to properties of real numbers. But NOT ALL properties of real numbers also hold for matrices. In particular, AB need not equal BA .

Theorem 8.3

The transpose of a product is the product of transposes in opposite order:

$$(AB)^T = B^T A^T$$

Definition 8.4

Let A be a square matrix. We write A^k for $A \cdots A$, k -times; that is A^k is obtained by multiplying A k -times with itself.

9. Elementary Matrices

Definition 9.1

An elementary matrix is one that is obtained by performing a single elementary row operation on an identity matrix.

A permutation matrix is one that is obtained by performing row exchanges on an identity matrix.

Theorem 9.2

If an elementary row operation is performed on an $m \times n$ -matrix A , the resulting matrix can be written as EA , where the $m \times m$ -matrix E is created by performing the same row operation on I_m .

Theorem 9.3

Let A, B be two $m \times n$ -matrices and row-equivalent. Then there is a sequence of $m \times m$ -elementary matrices $E_1, \dots, E_\ell$ such that

$$E_\ell \cdots E_1 A = B.$$

10. Inverse of a Matrix

Remark

The inverse of a real number a is denoted by a^{-1} . For example, $7^{-1} = 1/7$ and $7 \cdot 7^{-1} = 7^{-1} \cdot 7 = 1$. Note that not all real numbers have an inverse. Namely, 0^{-1} is not defined as there is no real number b such that $0 \cdot b = 1$.

Definition 10.1

An $n \times n$ matrix A is said to be invertible if there is an $n \times n$ matrix C satisfying

$$CA = AC = I_n$$

where I_n is the $n \times n$ identity matrix. We call C the inverse of A .

Theorem 10.2

Let A be an invertible matrix, then its inverse C is unique.

Theorem 10.3

Suppose A and B are invertible. Then:

1. A^{-1} is invertible and $(A^{-1})^{-1} = A$ (i.e. A is the inverse of A^{-1}).
2. AB is invertible and $(AB)^{-1} = B^{-1}A^{-1}$.
3. A^T is invertible and $(A^T)^{-1} = (A^{-1})^T$.

Theorem 10.4

Let A be an invertible $n \times n$ matrix. Then for each b in $\mathbb{R}^n$, the equation $Ax = b$ has the unique solution $x = A^{-1}b$.

11. Computing the Inverse

Theorem 11.1

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. If $ad - bc \neq 0$, then A is invertible and

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

If $ad - bc = 0$, then A is not invertible.

Theorem 11.2

Let A be an $n \times n$ -matrix. The following are equivalent:

- A is invertible.
- the reduced echelon form of A is I_n .

Theorem 11.3

Suppose A is invertible. Then every sequence of elementary row operations that reduces A to I_n will also transform I_n to A^{-1} .

Algorithm 11.4

- Place A and I side-by-side to form an augmented matrix $[A | I]$. This is an $n \times 2n$ matrix (Big Augmented Matrix), instead of $n \times (n + 1)$.
- Perform row operations on this matrix (which will produce identical operations on A and I).
- By Theorem: $[A | I]$ will row reduce to $[I | A^{-1}]$ or A is not invertible.

12. LU Decomposition

Definition 12.1

An $n \times n$ matrix A is called

- upper triangular if it is of the form

$$\begin{bmatrix} \star & \star & \star & \star & \star \\ 0 & \star & \star & \star & \star \\ 0 & 0 & \star & \star & \star \\ 0 & 0 & 0 & \ddots & \vdots \\ 0 & 0 & 0 & 0 & \star \end{bmatrix},$$

- lower triangular if it is of the form

$$\begin{bmatrix} \star & 0 & 0 & 0 & 0 \\ \star & \star & 0 & 0 & 0 \\ \star & \star & \star & 0 & 0 \\ \star & \star & \star & \ddots & \vdots \\ \star & \star & \star & \star & \star \end{bmatrix}.$$

Theorem 12.2

The product of two lower (upper) triangular matrices is lower (upper) triangular.

Remark

The inverse of a lower (upper) triangular matrix (if it exists) is again lower (upper) triangular.

Definition 12.3

A matrix A has an LU decomposition if there is a lower triangular matrix L and an upper triangular matrix U such that $A = LU$.

Theorem 12.4

Let A be an $n \times n$ -matrix. If A can be brought to echelon form just using row operations of the form $R_i \rightarrow R_i + cR_j$ where $j < i$, then A has an LU-decomposition.

Remark

Not every matrix has an LU decomposition. The LU decomposition of a matrix is not unique.

13. Solving Linear Systems using LU Decomposition

Theorem 13.1

Let A be an $n \times n$ -matrix such that $A = LU$, where L is a lower triangular matrix and U is an upper triangular matrix, and let $b \in \mathbb{R}^n$. In order to find a solution of the linear system

$$Ax = b,$$

it is enough to find a solution of the linear system

$$Ux = c,$$

where c satisfies $Lc = b$.

Theorem 13.2

Let A be an $n \times n$ matrix. Then there is a permutation matrix P such that PA has an LU-decomposition.

13.1. Spring-Mass Systems

See the notes for information about this application.

14. Inner Product and Orthogonality

Definition 14.1

The inner product of $v, w \in \mathbb{R}^n$ is

$$v \cdot w = v^T w.$$

Theorem 14.2

Let u, v and w be vectors in $\mathbb{R}^n$, and let c be any scalar. Then

1. $u \cdot v = v \cdot u$
2. $(u + v) \cdot w = u \cdot w + v \cdot w$
3. $(cu) \cdot v = c(u \cdot v) = u \cdot (cv)$
4. $u \cdot u \geq 0$, and $u \cdot u = 0$ if and only if $u = 0$.

Definition 14.3

Let $v, w \in \mathbb{R}^n$.

- The norm (or length) of v is

$$\|v\| = \sqrt{v \cdot v} = \sqrt{v_1^2 + \dots + v_n^2}.$$

- The distance between v and w is

$$\text{dist}(v, w) = \|v - w\|.$$

Definition 14.4

Let $v, w \in \mathbb{R}^n$. We say v and w are orthogonal if $v \cdot w = 0$.

Theorem 14.5

Let $v, w \in \mathbb{R}^n$ be non-zero. Then v and w are orthogonal if and only if they are perpendicular (i.e. if they form a right angle).

Definition 14.6

A set of vectors in $\mathbb{R}^n$ is pairwise orthogonal if each pairing of them is orthogonal. We call such a set an orthogonal set.

Definition 14.7

A unit vector in $\mathbb{R}^n$ is a vector of length 1.

Definition 14.8

A set of vectors forms an orthonormal set if this set is an orthogonal set and all vectors in the set are unit vectors.

15. Subspaces

Definition 15.1

A non-empty subset H of $\mathbb{R}^n$ is a subspace of $\mathbb{R}^n$ if it satisfies the following two conditions:

- if $u, v \in H$, then the sum $u + v \in H$. (H is closed under vector addition), and
- if $u \in H$ and c is a scalar, then $cu \in H$. (H is closed under scalar multiplication.)

Theorem 15.2

Let $v_1, v_2, \dots, v_m \in \mathbb{R}^n$. Then $\text{Span}(v_1, v_2, \dots, v_m)$ is a subspace of $\mathbb{R}^n$.

16. Column Spaces and Null Spaces

Definition 16.1

The column space, written as $\text{Col}(A)$, of an $m \times n$ matrix A is the set of all linear combinations of the columns of A . If $A = [a_1 \ a_2 \ \dots \ a_n]$, then $\text{Col}(A) = \text{Span}(a_1, a_2, \dots, a_n)$.

Theorem 16.2

Let A be an $m \times n$ matrix. Then $\text{Col}(A)$ is a subspace of $\mathbb{R}^m$.

Theorem 16.3

Let A be an $m \times n$ matrix and $b \in \mathbb{R}^m$. Then b is in $\text{Col}(A)$ if and only if the linear system $Ax = b$ has a solution.

Definition 16.4

The nullspace of an $m \times n$ matrix A , written as $\text{Nul}(A)$, is the set of all solutions to the homogeneous equation $Ax = 0$; that is, $\text{Nul}(A) = \{v \in \mathbb{R}^n : Av = 0\}$.

Theorem 16.5

The nullspace of an $m \times n$ matrix A is a subspace of $\mathbb{R}^n$.

Theorem 16.6

Let A be an $m \times n$ matrix, let $b \in \mathbb{R}^m$, and let $w \in \mathbb{R}^n$ such that $Aw = b$. Then $\{v \in \mathbb{R}^n : Av = b\} = w + \text{Nul}(A)$.

17. Abstract Vector Spaces

Definition 17.1

A vector space is a non-empty set V of objects, called vectors, for which linear combinations make

sense. More precisely, there are two operations on V , called addition and multiplication by scalars (real numbers), satisfying the following axioms for all $u, v, w \in V$ and for all scalars $c, d \in \mathbb{R}$:

- $u + v$ is in V . (V is “closed under addition”.)
- $u + v = v + u$.
- $(u + v) + w = u + (v + w)$.
- There is a vector (called the zero vector) 0_V in V such that $u + 0_V = u$.
- For each u in V , there is a vector $-u$ in V satisfying $u + (-u) = 0_V$.
- cu is in V . (V is “closed under scalar multiplication”.)
- $c(u + v) = cu + cv$.
- $(c + d)u = cu + du$.
- $(cd)u = c(du)$.
- $1u = u$.

Definition 17.2

Let V be a vector space. A non-empty subset $W \subseteq V$ is a subspace of V if

- $u + v$ for all $u, v \in U$ (closed under addition)
- cu for all $u \in U$ and $c \in \mathbb{R}$ (closed under scalar multiplication)

18. Linear Independence

Definition 18.1

Vectors $v_1, \dots, v_p$ are said to be linearly independent if the equation

$$x_1 v_1 + x_2 v_2 + \dots + x_p v_p = 0$$

has only the trivial solution (namely, $x_1 = x_2 = \dots = x_p = 0$). We say the vectors are linearly dependent if they are not linearly independent.

Theorem 18.2

Vectors $v_1, \dots, v_p$ are linearly dependent if and only if there is $i \in \{1, \dots, p\}$ such that $v_i \in \text{Span}(v_1, \dots, v_{i-1}, v_{i+1}, \dots, v_p)$.

Theorem 18.3

Let A be an $m \times n$ matrix. The following are equivalent:

- The columns of A are linearly independent.
- $Ax = 0$ has only the solution $x = 0$.
- A has n pivots.
- there are no free variables for $Ax = 0$.

19. Bases and Dimension

Definition 19.1

Let V be a vector space. A sequence of vectors $(v_1, \dots, v_p)$ in V is a basis of V if

- $V = \text{Span}(v_1, \dots, v_p)$, and
- $(v_1, \dots, v_p)$ are linearly independent.

Theorem 19.2

Every two bases in a vector space V contain the same number of vectors.

Definition 19.3

The number of vectors in a basis of V is the dimension of V .

Theorem 19.4

Suppose that V has dimension d . Then

- A sequence of d vectors in V is a basis if they span V .
- A sequence of d vectors in V is a basis if they are linearly independent.

Theorem 19.5

A basis is a minimal spanning set of V ; that is the elements of the basis span V but you cannot delete any of these elements and still get all of V .

20. The Four Fundamental Subspaces

Algorithm 20.1

To find a basis for $\text{Nul}(A)$:

- Find the parametric form of the solutions to $Ax = 0$.
- Express solutions x as a linear combination of vectors with the free variables as coefficients.
- Use these vectors as a basis of $\text{Nul}(A)$.

Definition 20.2

The rank of a matrix is the number of pivots it has.

Theorem 20.3

Let A be an $m \times n$ matrix with rank r . Then $\dim \text{Nul}(A) = n - r$.

Theorem 20.4

Let A be an $m \times n$ matrix with rank r . The pivot columns of A form a basis of $\text{Col}(A)$. In particular, $\dim \text{Col}(A) = r$.

Theorem 20.5

Let A, B be two row-equivalent matrices. Then $\text{Col}(A^T) = \text{Col}(B^T)$.

Theorem 20.6

Let A be an $m \times n$ matrix with rank r . Then the non-zero rows in an echelon form of A form a basis of $\text{Col}(A^T)$, and thus $\dim(A^T) = r$.

Theorem 20.7

Let A be an $m \times n$ matrix with rank r . Then

- $\dim \text{Col}(A) = \dim \text{Col}(A^T) = r$, and
- $\dim \text{Nul}(A) = n - r$.

20.1. Graphs

Definition 20.1.1

A graph is a set of nodes (or: vertices) that are connected through edges.

Definition 20.1.2

Let G be a graph with n nodes. The adjacency matrix of G is the $n \times n$ -matrix $A = (a_{ij})$ such that

$$a_{ij} = \begin{cases} 1 & \text{if there is an edge between node } i \text{ and node } j \\ 0 & \text{otherwise} \end{cases}$$

Definition 20.1.3

A walk of length k on a graph is a sequence of $k + 1$ vertices and k edges between two nodes (including the start and end) that may repeat. A path is a walk in which all vertices are distinct.

Definition 20.1.4

A graph is connected if for every pair of nodes i and j there is a walk from node i to node j . A graph is disconnected if it is not connected.

Theorem 20.1.5

Let G be a graph and let A be its adjacency matrix. Then the entry in the i -th row and j -th column of A^ℓ is the number of walks of length ℓ from node j to node i on G .

Definition 20.1.6

A directed graph is a set of vertices connected by edges, where the edges have a direction associated with them.

Definition 20.1.7

Let G be a directed graph with m edges and n nodes. The adjacency matrix of G is the $n \times n$ matrix $A = (a_{ij})_{i,j}$ with

$$a_{i,j} = \begin{cases} 1 & \text{if there is a directed edge from node } j \text{ to node } i \\ 0 & \text{otherwise} \end{cases}$$

Definition 20.1.8

Let G be a directed graph with m edges and n nodes. The edge-node incidence matrix of G is the $m \times n$ matrix $A = (a_{ij})_{i,j}$ with

$$a_{i,j} = \begin{cases} -1 & \text{if edge } i \text{ leaves node } j \\ +1 & \text{if edge } i \text{ enters node } j \\ 0 & \text{otherwise} \end{cases}$$

Definition 20.1.9

A connected component of an undirected graph is a part in which any two vertices are connected to each other by paths, and which is connected to no additional vertices in the rest of the graph. The connected components of a directed graph are those of its underlying undirected graph. A graph is connected if it has only one connected component.

Theorem 20.1.10

Let G be a directed graph and let A be its edge-node incidence matrix. Then $\dim \text{Nul}(A)$ is equal to the number of connected components of G .

Definition 20.1.11

A cycle in an undirected graph is a path in which all edges are distinct and the only repeated vertices are the first and last vertices. By cycles of a directed graph we mean those of its underlying undirected graph.

Definition 20.1.12

The span of all cycle vectors of a graph G is called the cycle space.

Theorem 20.1.13

Let G be a directed graph and let A be its edge-node incidence matrix. Then the cycle space of G is equal to $\text{Nul}(A^T)$.

21. Orthogonal Complements

Definition 21.1

Let W be a subspace of $\mathbb{R}^n$. The orthogonal complement of W is the subspace $W^\perp$ of all vectors that are orthogonal to W ; that is

$$W^\perp := \{v \in \mathbb{R}^n : v \cdot w = 0 \text{ for all } w \in W\}.$$

 Remark

Observe that $(W^\perp)^\perp = W$.

Theorem 21.2

Let A be an $m \times n$ matrix. Then $\text{Nul}(A)$ is the orthogonal complement of $\text{Col}(A^T)$; that is $\text{Nul}(A) = \text{Col}(A^T)^\perp$.

 Remark

It follows that

$$\bullet \text{Nul}(A)^\perp = \text{Col}(A^T).$$

$$\bullet \text{Nul}(A^T) = \text{Col}(A)^\perp.$$

Theorem 21.3

Let V be a subspace of $\mathbb{R}^n$. Then $\dim V + \dim V^\perp = n$.

22. Coordinates

Theorem 22.1

Let $(v_1, \dots, v_p)$ be a basis of V . Then every vector w in V can be expressed uniquely as

$$w = c_1 v_1 + \dots + c_p v_p.$$

Definition 22.2

Let $B = (v_1, v_2, \dots, v_p)$ be an (ordered) basis of V , and let $w \in V$. The coordinate vector w_B of w with respect to the basis B is

$$w_B = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ \vdots \\ c_p \end{bmatrix}, \quad \text{if } w = c_1 v_1 + c_2 v_2 + \dots + c_p v_p.$$

Definition 22.3

In $\mathbb{R}^n$ let e_i denote the vector with a 1 in the i -th coordinate and 0's elsewhere. The standard basis of $\mathbb{R}^n$ is the ordered basis $E_n := (e_1, \dots, e_n)$.

Definition 22.4

Let B and C be two bases of $\mathbb{R}^n$. The change of basis matrix $I_{C,B}$ is the matrix such that for all $v \in \mathbb{R}^n$

$$I_{C,B} v_B = v_C$$

Theorem 22.5

Let $B = (b_1, \dots, b_n)$ be a basis of $\mathbb{R}^n$. Then

$$I_{E_n, B} = [b_1 \ \dots \ b_n]$$

That is, for all $v \in \mathbb{R}^n$,

$$v = [b_1 \ \dots \ b_n] v_B.$$

 Remark

Let B and C be bases in $\mathbb{R}^n$.

• To compute I_{B, E_n} , notice that

$$v = I_{E_n, B} v_B \rightsquigarrow I_{E_n, B}^{-1} v = v_B \rightsquigarrow I_{B, E_n} = I_{E_n, B}^{-1}$$

• To compute $I_{B, C}$, we can either use

$$I_{B, E_n} I_{E_n, C} v_C = I_{B, E_n} v = v_B \rightsquigarrow I_{B, C} = I_{B, E_n} I_{E_n, C}$$

or use

$$I_{C, B} = [(b_1)_C \ \dots \ (b_n)_C]$$

23. Orthogonal and Orthonormal Bases

Theorem 23.1

Let $v_1, \dots, v_m \in \mathbb{R}^n$ be non-zero and pairwise orthogonal. Then $v_1, \dots, v_m$ are linearly independent.

 Remark

The theorem implies that a set of n orthonormal vectors in $\mathbb{R}^n$ is a basis of $\mathbb{R}^n$.

Definition 23.2

An orthogonal basis (an orthonormal basis) is an orthogonal set of vectors (an orthonormal set of vectors) that forms a basis.

Remark

When B is orthonormal, then $b_i \cdot b_i = 1$ for $i = 1, \dots, n$.

Theorem 23.3

Let $B := (b_1, b_2, \dots, b_n)$ be an orthogonal basis of $\mathbb{R}^n$, and let $v \in \mathbb{R}^n$. Then

$$v = \frac{v \cdot b_1}{b_1 \cdot b_1} b_1 + \dots + \frac{v \cdot b_n}{b_n \cdot b_n} b_n.$$

Theorem 23.4

Let $U = (u_1, \dots, u_n)$ be an orthonormal basis of $\mathbb{R}^n$. Then

$$I_{U, E_n} = [u_1 \ \dots \ u_n]^T.$$

Definition 23.5

An $n \times n$ -matrix Q is orthogonal if $Q^{-1} = Q^T$.

24. Linear Transformations

Definition 24.1

Let V and W be vector spaces. A map $T : V \rightarrow W$ is a linear transformation if

$$T(av + bw) = aT(v) + bT(w)$$

for all $v, w \in V$ and all $a, b \in \mathbb{R}$.

Remark

$$\begin{aligned} T(0_V) &= T(0 \cdot 0_V) = 0 \cdot T(0_V) = 0_W \leadsto \\ T(0_V) &= 0_W \end{aligned}$$

Theorem 24.2

Let V, W be two vector spaces, let $T : V \rightarrow W$ be a linear transformation and let $(v_1, \dots, v_n)$ be a basis of V . Then T is completely determined by the values $T(v_1), \dots, T(v_n)$.

Theorem 24.3

Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation. Then there is an $m \times n$ matrix A such that

- $T(v) = Av$, for all $v \in \mathbb{R}^n$.
- $A = [T(e_1) \ T(e_2) \ \dots \ T(e_n)]$, where $(e_1, e_2, \dots, e_n)$ is the standard basis of $\mathbb{R}^n$.

Remark

We call this A the coordinate matrix of T with respect to the standard bases – we write T_{E_m, E_n} .

25. Coordinate Matrices of Linear Transformations

Theorem 25.1

Let V, W be two vector spaces, let $B = (b_1, \dots, b_n)$ be a basis of V and $C = (c_1, \dots, c_m)$ be a basis of W , and let $T : V \rightarrow W$ be a linear transformation. Then there is an $m \times n$ matrix $T_{C, B}$ such that

- $T(v)_C = T_{C, B} v_B$, for all $v \in V$.
- $T_{C, B} = [T(b_1)_C \ T(b_2)_C \ \dots \ T(b_n)_C]$.

Theorem 25.2

Let $T : \mathbb{R}^m \rightarrow \mathbb{R}^n$ be a linear transformation and A and B be two bases of $\mathbb{R}^m$ and C, D be two bases of $\mathbb{R}^n$. Then

$$T_{C, A} = I_{C, D} T_{D, B} I_{B, A}.$$

26. Determinants

Goal. Define a determinant function that takes an $n \times n$ matrix A to a real number $\det(A)$, such that $\det(A) \neq 0$ if and only if A is invertible. We start with small matrices:

Definition 26.1

The determinant of

- a 1×1 matrix is $\det\begin{pmatrix} a \end{pmatrix} = a$.
- a 2×2 matrix is $\det\begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$.

Remark

We will write both $\det\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and $\begin{vmatrix} a & b \\ c & d \end{vmatrix}$ for the determinant.

Definition 26.2

The determinant is the function that assigns to each $n \times n$ -matrix a number and satisfies the following conditions:

- (Normalization) $\det I_n = 1$,
- It is affected by elementary row operations as follows:
 - ▶ (Replacement) Adding a multiple of one row to another row does not change the determinant.
 - ▶ (Interchange) Interchanging two different rows reverses the sign of the determinant.
 - ▶ (Scaling) Multiplying all entries in a row by s , multiplies the determinant by s .

Theorem 26.3

The determinant of a triangular matrix is the product of the diagonal entries.

Theorem 26.4

Let A be an $n \times n$ -matrix, then $\det(A) = 0$ if and only if A is not invertible.

Theorem 26.5

Let A, B be two $n \times n$ -matrices. Then $\det(AB) = \det(A)\det(B)$.

Theorem 26.6

Let A be an $n \times n$ -matrix. Then $\det(A^T) = \det(A)$.

27. Cofactor expansion

Remark

Let A be an $n \times n$ -matrix. We denote by A_{ij} the matrix obtained from matrix A by deleting the i -th row and j -th column of A .

Definition 27.1

Let A be an $n \times n$ -matrix. The (i, j) -cofactor of A is the scalar C_{ij} defined by

$$C_{ij} = (-1)^{i+j} \det A_{ij}.$$

Theorem 27.2

Let A be an $n \times n$ -matrix. Then for every $i, j \in \{1, \dots, n\}$

$$\det A = a_{i1}C_{i1} + a_{i2}C_{i2} + \dots + a_{in}C_{in} \quad (\text{expansion across row } i) \\ = a_{1j}C_{1j} + a_{2j}C_{2j} + \dots + a_{nj}C_{nj} \quad (\text{expansion down column } j)$$

28. Eigenvectors

Definition 28.1

Let A be an $n \times n$ matrix. An eigenvector of A is a nonzero $v \in \mathbb{R}^n$ such that $Av = \lambda v$, where λ is a scalar, known as the eigenvalue associated with v .

Definition 28.2

Let λ be an eigenvalue of A . The eigenspace of A associated with λ is the set

$$\text{Eig}_\lambda(A) := \{v : Av = \lambda v\}.$$

It consists of all the eigenvectors of A with eigenvalue λ and the zero vector.

29. Computing Eigenvalues and Eigenvectors

Theorem 29.1

Let A be an $n \times n$ matrix and λ be a scalar. Then λ is an eigenvalue of A if and only if $\det(A - \lambda I) = 0$.

Theorem 29.2

Let A be an $n \times n$ matrix. Then $p_A(t) := \det(A - tI)$ is a polynomial of degree n . Thus A has at most n eigenvalues.

Definition 29.3

We call $p_A(t)$ the characteristic polynomial of A .

Theorem 29.4

Let A be an $n \times n$ matrix and let λ be an eigenvalue of A . Then

$$\text{Eig}_\lambda(A) = \text{Nul}(A - \lambda I).$$

30. Properties of Eigenvectors and Eigenvalues

Definition 30.1

Let A be an $n \times n$ matrix and let λ be an eigenvalue of A .

- The algebraic multiplicity of λ is its multiplicity as a root of the characteristic polynomial, that is, the largest integer k such that $(t - \lambda)^k$ divides $p_A(t)$.
- The geometric multiplicity of λ is the dimension of the eigenspace $\text{Eig}_\lambda(A)$ of λ .

Theorem 30.2

Let A be an $n \times n$ matrix and let $v_1, \dots, v_m$ be eigenvectors of A corresponding to different eigenvalues. Then $v_1, \dots, v_m$ are linearly independent.

Definition 30.3

Let

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix}$$

The trace of A is the sum of the diagonal entries of A ; that is

$$\text{tr}(A) = a_{11} + a_{22} + \dots + a_{nn}.$$

Theorem 30.4

Let A be an $n \times n$ matrix with eigenvalues $\lambda_1, \dots, \lambda_n$. Then

- $\text{tr}(A) = \lambda_1 + \lambda_2 + \dots + \lambda_n$.
- $\det(A) = \lambda_1 \cdot \lambda_2 \cdot \dots \cdot \lambda_n$.

Theorem 30.5

Let A be a 2×2 -matrix. Then the characteristic polynomial of A is

$$p(\lambda) = \lambda^2 - \operatorname{tr}(A)\lambda + \det(A).$$

31. Markov matrices

Definition 31.1

An $n \times n$ matrix A is a Markov matrix (or: stochastic matrix) if it has only non-negative entries, and the entries in each column add up to 1.

A vector in $\mathbb{R}^n$ is a probability vector (or: stochastic vector) if it has only non-negative entries, and the entries add up to 1.

Theorem 31.2

Let A be a Markov matrix. Then

- 1 is an eigenvalue of A and every other eigenvalue λ of A satisfies $|\lambda| \leq 1$.
- If A has only positive entries, then any other eigenvalue satisfies $|\lambda| < 1$.

Definition 31.3

A stationary probability vector of a Markov matrix is a probability vector v that is an eigenvector of A corresponding to the eigenvalue 1.

Theorem 31.4

Let A be an $n \times n$ -Markov matrix with only positive entries and let $z \in \mathbb{R}^n$ be a probability vector. Then

$$z_\infty := \lim_{k \rightarrow \infty} A^k z \text{ exists,}$$

and z_∞ is a stationary probability vector of A (i.e. $Az_\infty = z_\infty$).

32. Diagonalization

Definition 32.1

A square matrix A is said to be diagonalizable if there is an invertible matrix P and a diagonal matrix D such that $A = PDP^{-1}$.

Theorem 32.2

Let A be an $n \times n$ matrix that has n linearly independent eigenvectors $v_1, v_2, \dots, v_n$ with associated eigenvalues $\lambda_1, \dots, \lambda_n$. Then A is diagonalizable as PDP^{-1} , where

$$P = [v_1 \ \dots \ v_n] \quad \text{and} \quad D = \begin{bmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_n \end{bmatrix}$$

Definition 32.3

Vectors $v_1, \dots, v_n$ form an eigenbasis of an $n \times n$ matrix A if $v_1, \dots, v_n$ form a basis of $\mathbb{R}^n$ and $v_1, \dots, v_n$ are all eigenvectors of A .

Theorem 32.4

The following are equivalent for an $n \times n$ matrix A :

- A has an eigenbasis.
- A is diagonalizable.
- The geometric multiplicities of all eigenvalues of A sum up to n .

Theorem 32.5

Let A be an $n \times n$ matrix and let $B = (v_1, \dots, v_n)$ be an eigenbasis of A . Then there is a diagonal matrix D such that

$$A = I_{E_n, B} D I_{B, E_n}.$$

33. Powers of Matrices

Key idea. If a matrix A has an eigenbasis, then it is easy to compute A^k , even for large k .

Theorem 33.1

If $A = PDP^{-1}$, where D is a diagonal matrix, then for any m ,

$$A^m = PD^m P^{-1}.$$

34. Matrix Exponential

Definition 34.1

Let A be an $n \times n$ -matrix. We define the matrix exponential e^{At} as

$$e^{At} = \sum_{k \geq 0} \frac{(At)^k}{k!} = I + At + \frac{(At)^2}{2!} + \frac{(At)^3}{3!} + \dots$$

Theorem 34.2

Let A be an $n \times n$ matrix. Then

- the series e^{At} always converges,
- $e^{At} e^{-At} = I_n$,
- $e^{At} e^{As} = e^{A(t+s)}$,
- $\frac{d}{dt}(e^{At}) = Ae^{At}$.

Theorem 34.3

Let A be an $n \times n$ matrix such that $A = PDP^{-1}$ for some invertible matrix P and some diagonal matrix D . Then

$$e^{At} = P e^{Dt} P^{-1}.$$

34.1. Linear Differential Equations

Definition 34.1.1

A linear (first order) differential equation is an equation of the form

$$\frac{du}{dt} = Au$$

where u is a function from $\mathbb{R}$ to $\mathbb{R}^n$. A further condition $u(0) = v$, for some v in $\mathbb{R}^n$ is called an initial condition.

Theorem 34.1.2

Let A be an $n \times n$ matrix and $v \in \mathbb{R}^n$. The solution of the differential equation $\frac{du}{dt} = Au$ with initial condition $u(0) = v$ is $u(t) = e^{At}v$.

Theorem 34.1.3

Let A be an $n \times n$ -matrix, and let $v \in \mathbb{R}^n$ be an eigenvector of A with eigenvalue λ , then $e^{At}v = e^{\lambda t}v$.

Theorem 34.1.4

Let A be an $n \times n$ -matrix and $(v_1, \dots, v_n)$ be an eigenbasis of A with eigenvalues $\lambda_1, \dots, \lambda_n$. If $v = c_1v_1 + \dots + c_nv_n$, then the unique solution to the differential equation $\frac{du}{dt} = Au$ with initial condition $u(0) = v$ is

$$e^{At}v = c_1e^{\lambda_1 t}v_1 + \dots + c_ne^{\lambda_n t}v_n.$$

35. Orthogonal Projections Onto a Line

Definition 35.1

Let $v, w \in \mathbb{R}^n$. The orthogonal projection of v onto the line spanned by w is

$$\text{proj}_w v := \frac{w \cdot v}{w \cdot w} w.$$

Theorem 35.2

Let $v, w \in \mathbb{R}^n$. Then $\text{proj}_w v$ is the point in $\text{Span}(w)$ closest to v ; that is

$$\text{dist}(v, \text{proj}_w v) = \min_{u \in \text{Span}(w)} \text{dist}(v, u).$$

Remark

Note that $v - \text{proj}_w v$ (called the error term) is in $\text{Span}(w)^\perp$.

$$v = \underbrace{\text{proj}_w v}_{\in \text{Span}(w)} + \underbrace{v - \text{proj}_w v}_{\in \text{Span}(w)^\perp}$$

Theorem 35.3

Let $w \in \mathbb{R}^n$, then for all $v \in \mathbb{R}^n$

$$\text{proj}_w v = \left(\frac{1}{w \cdot w} ww^T \right) v.$$

Remark

Note that $\left(\frac{1}{w \cdot w} ww^T \right)$ is an $n \times n$ matrix; we call it the orthogonal projection matrix onto $\text{Span}(w)$.

36. Orthogonal Projection Onto a Subspace

Theorem 36.1

Let W be a subspace of $\mathbb{R}^n$ and $v \in \mathbb{R}^n$. Then v can be written uniquely as

$$v = \underbrace{\hat{v}}_{\text{in } W} + \underbrace{v^\perp}_{\text{in } W^\perp}$$

Definition 36.2

We say $\hat{v}$ is the orthogonal projection of v onto W – written $\text{proj}_W v$.

Remark

If $(w_1, \dots, w_m)$ is an orthogonal basis of W , then

$$\text{proj}_W v = \left(\frac{v \cdot w_1}{w_1 \cdot w_1} \right) w_1 + \dots + \left(\frac{v \cdot w_m}{w_m \cdot w_m} \right) w_m.$$

Theorem 36.3

The projection map $\text{proj}_W : \mathbb{R}^n \rightarrow \mathbb{R}^n$ that sends v to $\text{proj}_W v$ is linear.

Definition 36.4

Let matrix P_W be the matrix $(\text{proj}_W)_{E_n, E_n}$ that represents proj_W with respect to the standard basis; we call P_W the orthogonal projection matrix onto W .

37. Least Squares Solutions

Goal. Suppose $Ax = b$ is inconsistent. Can we still find something like a best solution?

Definition 37.1

Let A be an $m \times n$ matrix and $b \in \mathbb{R}^m$. A least squares solution (short: LSQ solution) of the system $Ax = b$ is a vector $\hat{x} \in \mathbb{R}^n$ such that

$$\text{dist}(A\hat{x}, b) = \min_{x \in \mathbb{R}^n} \text{dist}(Ax, b).$$

Theorem 37.2

Let A be an $m \times n$ matrix and $b \in \mathbb{R}^m$. Then $\hat{x}$ is an LSQ solution to $Ax = b$ if and only if $A\hat{x} = \text{proj}_{\text{Col}(A)} b$.

Theorem 37.3

Let A be an $m \times n$ matrix and $b \in \mathbb{R}^m$. Then $\hat{x}$ is an LSQ solution to $Ax = b$ if and only if $A^T A\hat{x} = A^T b$.

Theorem 37.4

Let A be an $m \times n$ matrix with linearly independent columns and $b \in \mathbb{R}^m$. Then

$$\text{proj}_{\text{Col}(A)} b = A(A^T A)^{-1} A^T b.$$

38. Linear Regression

There are no definitions or theorems in this module, see the lecture notes for examples.

39. Gram-Schmidt Method

Theorem 39.1

Every subspace of $\mathbb{R}^n$ has an orthonormal basis.

Algorithm 39.2 (Gram-Schmidt orthonormalization)

Given a basis $a_1, \dots, a_m$, we produce an orthogonal basis $b_1, \dots, b_m$ and an orthonormal basis $q_1, \dots, q_m$.

$$b_1 = a_1, \quad q_1 = \frac{b_1}{\|b_1\|}$$

$$b_2 = a_2 - \underbrace{\text{proj}_{\text{Span}(q_1)} a_2}_{=(a_2 \cdot q_1) q_1}, \quad q_2 = \frac{b_2}{\|b_2\|}$$

$$b_3 = a_3 - \underbrace{\text{proj}_{\text{Span}(q_1, q_2)} a_3}_{(a_3 \cdot q_1) q_1 + (a_3 \cdot q_2) q_2}, \quad q_3 = \frac{b_3}{\|b_3\|}$$

...

Remark

Notice that:

- $\text{Span}(q_1, \dots, q_i) = \text{Span}(a_1, \dots, a_i)$ for $i = 1, \dots, m$, and
- $q_j \notin \text{Span}(a_1, \dots, a_i)$ for all $j > i$.

Theorem 39.3 (QR decomposition)

Let A be an $m \times n$ matrix of rank n . There is an $m \times n$ -matrix Q with orthonormal columns and an

upper triangular $n \times n$ invertible matrix R such that $A = QR$.

40. The Spectral Theorem

Theorem 40.1

Let A be a symmetric $n \times n$ matrix, then A has an orthonormal basis of eigenvectors.

Theorem 40.2

Let A be a symmetric $n \times n$ matrix. Then there is a diagonal matrix D and a matrix Q with orthonormal columns such that $A = QDQ^T$.

41. Singular Value Decomposition

Definition 41.1

Let A be an $m \times n$ matrix. A singular value decomposition of A is a decomposition $A = U\Sigma V^T$ where

- U is an $m \times m$ matrix with orthonormal columns,
- Σ is an $m \times n$ rectangular diagonal matrix with non-negative numbers on the diagonal,
- V is an $n \times n$ matrix with orthonormal columns.

Remark

The diagonal entries $\sigma_i = \Sigma_{ii}$ which are positive are called the singular values of A . We usually arrange them in decreasing order, that is

$$\sigma_1 \geq \sigma_2 \geq \dots$$

Remark

Let A be an $m \times n$ matrix with rank r . Recall that

- $\text{Nul}(A^T A) = \text{Nul}(A)$ and $\text{Nul}(AA^T) = \text{Nul}(A^T)$.
- $A^T A$ is symmetric and has rank r .

Algorithm 41.2

Let A be an $m \times n$ matrix with rank r .

- Find orthonormal eigenbasis $(v_1, \dots, v_n)$ of $A^T A$ with eigenvalues $\lambda_1 \geq \dots \geq \lambda_r > \lambda_{r+1} = 0 = \dots = \lambda_n$.
- Set $\sigma_i = \sqrt{\lambda_i}$ for $i = 1, \dots, n$.
- Set $u_1 = \frac{1}{\sigma_1} A v_1, \dots, u_r = \frac{1}{\sigma_r} A v_r$. (Magic: orthonormal!)
- Find $u_{r+1}, \dots, u_m \in \mathbb{R}^m$ such that $(u_1, \dots, u_m)$ is an orthonormal basis of $\mathbb{R}^m$.
- Set

$$U = [u_1 \ \dots \ u_m], \quad \Sigma = \begin{bmatrix} \sigma_1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \sigma_{\min\{m,n\}} \end{bmatrix}, \quad V = [v_1 \ \dots \ v_n]$$

Notice that $v_{r+1}, \dots, v_n \in \text{Nul}(A^T A)$, and hence $v_{r+1}, \dots, v_n \in \text{Nul}(A)$.

$$[v_1 \ \dots \ v_n] = [A v_1 \ \dots \ A v_n] = [\sigma_1 u_1 \ \dots \ \sigma_r u_r \ 0 \ \dots \ 0] = U \Sigma$$

Theorem 41.3

Let A be an $m \times n$ matrix with rank r , and suppose $U = [u_1 \ \dots \ u_m]$, $V = [v_1 \ \dots \ v_n]$, and Σ are such that $A = U\Sigma V^T$ is an SVD of A . Then

- $(u_1, \dots, u_r)$ is a basis of $\text{Col}(A)$.
- $(u_{r+1}, \dots, u_m)$ is a basis of $\text{Nul}(A^T)$.
- $(v_1, \dots, v_r)$ is a basis of $\text{Col}(A^T)$.
- $(v_{r+1}, \dots, v_n)$ is a basis of $\text{Nul}(A)$.

42. Low Rank Approximation

Theorem 42.1

Let A be an $m \times n$ matrix with rank r , and let $U = [u_1 \ \dots \ u_m]$, $V = [v_1 \ \dots \ v_n]$ be matrices with orthonormal columns and Σ be a rectangular diagonal $m \times n$ matrix such that $A = U\Sigma V^T$ is an SVD of A . Then

$$u_1 v_1^T + \sigma_2 u_2 v_2^T + \cdots + \sigma_r u_r v_r^T = [u_1 \cdots u_r] \begin{bmatrix} \sigma_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \sigma_r \end{bmatrix} [v_1 \cdots v_r]^T$$

Definition 42.2

For $k \leq r$, define

$$A_k = \sigma_1 u_1 v_1^T + \sigma_2 u_2 v_2^T + \cdots + \sigma_k u_k v_k^T.$$

Idea. If $\sigma_1 \gg \sigma_2 \gg \cdots$, then A_k is a good approximation of A .

Definition 42.3

Let A be an $m \times n$ matrix with rank r . A compact singular value decomposition of A is a decomposition $A = U_c \Sigma_c V_c^T$ where

- $U_c = [u_1 \cdots u_r]$ is an $m \times r$ matrix with orthonormal columns,
- Σ_c is an $r \times r$ diagonal matrix with positive diagonal elements,
- $V_c = [v_1 \cdots v_r]$ is an $n \times r$ matrix with orthonormal columns.

43. Pseudoinverses

Definition 43.1

Let A be an $m \times n$ matrix with rank r . Given the compact singular value decomposition $A = U_c \Sigma_c V_c^T$, where

- $U_c = [u_1 \cdots u_r]$ is an $m \times r$ matrix with orthonormal columns,
- Σ_c is an $r \times r$ diagonal matrix with positive diagonal elements,
- $V_c = [v_1 \cdots v_r]$ is an $n \times r$ matrix with orthonormal columns,

we define the pseudoinverse A^+ of A as $V_c \Sigma_c^{-1} U_c^T$.

Theorem 43.2

Let $b \in \text{Col}(A^T)$ and $w \in \text{Col}(A)$. Then $A^+ A v = v$ and $A A^+ w = w$.

Remark

If A is $n \times n$ and invertible, then $\text{Col}(A) = \mathbb{R}^n$. Thus $A^{-1} = A^+$.

Theorem 43.3

Let A be an $m \times n$ matrix and let $b \in \mathbb{R}^m$. Then $A^+ b$ is the LSQ solution of $Ax = b$ (with minimum length).

Remark

This is particularly useful, when solving many different LSQ problems of the form $Ax = b$, where A stays the same, but b varies.

44. Principal Component Analysis

Setup.

- Given m objects, we measure the same n variables.
- Thus m samples of n -dimensional data $\leadsto m \times n$ matrix (each row is a sample)
- Analyse this matrix to understand what drives the variance in the data.

Definition 44.1

Let $X = [a_1 \cdots a_m]^T$ be an $m \times n$ matrix. We define the column average $\mu(X)$ of X as

$$\mu(X) := \frac{1}{m}(a_1 + \cdots + a_m).$$

We say X is centered if $\mu(X) = 0$, and for X centered, we define the covariance matrix $\text{cov}(X)$ of X as $\frac{1}{m-1} X^T X$.

Principal component analysis (PCA):

- Input: centered $m \times n$ -matrix X .
- Compute $\text{cov}(X)$.
- Since $\text{cov}(X)$ is symmetric, we can find an orthonormal eigenbasis $v_1, v_2, \dots, v_n$ of $\text{cov}(X)$ with eigenvalues $\lambda_1 \geq \cdots \geq \lambda_n \geq 0$.
- Write $\text{cov}(X)$ as a sum of rank 1 matrices:

$$\text{cov}(X) = \lambda_1 v_1 v_1^T + \cdots + \lambda_n v_n v_n^T.$$

- Each principal component v_i explains part of the variance of the data. The larger λ_i , the more of the variance is explained by v_i .

PCA using SVD

- Let X be a centered data matrix. Observe that $X^T X = (m-1) \text{cov}(X)$.
- To find an orthonormal eigenbasis of $\text{cov}(X)$, it is enough to find an orthonormal eigenbasis of $X^T X$.
- Compute the SVD of $X \leadsto X = U \Sigma V^T$.
- The columns of $V = [v_1 \cdots v_n]$ are the desired orthonormal eigenbasis.
- If σ_i is the singular value for v_i , then

$$\lambda_i = \frac{\sigma_i^2}{m-1}$$

45. Review of Complex Numbers

Definition 45.1

$\mathbb{C} = \{x + iy \mid x, y \in \mathbb{R}\}$ where $i = \sqrt{-1}$, or $i^2 = -1$.

- The real part of z , denoted $\text{Re}(z)$ is defined by $\text{Re}(z) = x$.
- The imaginary part of z , denoted $\text{Im}(z)$ is defined by $\text{Im}(z) = y$.
- The complex conjugate of z , denoted $\bar{z}$, is defined by $\bar{z} = x - iy$.

- The absolute value, or magnitude of z , denoted $|z|$ or $\|z\|$, is given by

$$|z| = \sqrt{x^2 + y^2}.$$

Remark

Any point in $\mathbb{R}^2$ can be viewed as a complex number: $\begin{bmatrix} x \\ y \end{bmatrix} \leftrightarrow x + iy$.

Definition 45.2

Given $z = x + iy$, $w = u + iv$, we define $z + w = (x + u) + i(y + v)$, and

$$\begin{aligned} zw &= (x + iy)(u + iv) \\ &= xu + x(iv) + (iy)u + (iy)(iv) \\ &= (xu - yv) + i(xv + yu) \end{aligned}$$

Theorem 45.3

Let $z \in \mathbb{C}$.

- $\overline{\overline{z}} = z$
- $|z|^2 = z\overline{z}$
- $|\overline{z}| = |z|$

46. Complex Linear Algebra

Goal. Use complex numbers (instead of real numbers) as scalars.

Definition 46.1

The (complex) vector space $\mathbb{C}^n$ is the set of all

complex column vectors $z = \begin{bmatrix} z_1 \\ z_2 \\ \vdots \\ z_n \end{bmatrix}$, where

$z_1, z_2, \dots, z_n$ are complex numbers.

Remark

- Now multiplication by a complex scalar makes sense.
- We can define subspaces, Span, independence, basis, dimension for $\mathbb{C}^n$ in the usual way.
- We can multiply complex vectors by complex matrices. Column space and Null space still make sense.
- The only difference is the dot product, you need to use the complex conjugate to get a good notion of length:

$$\begin{bmatrix} z_1 \\ \vdots \\ z_n \end{bmatrix} \cdot \begin{bmatrix} w_1 \\ \vdots \\ w_n \end{bmatrix} = z_1 \overline{w_1} + z_2 \overline{w_2} + \dots + z_n \overline{w_n}.$$

Definition 46.2

Let A be an $m \times n$ -matrix. The conjugate matrix $\overline{A}$ of A is obtained from A by taking the complex conjugate of each entry of A .

Theorem 46.3

Let A be a matrix with real entries and λ is an eigenvalue of A . Then $\overline{\lambda}$ is also an eigenvalue. Furthermore, if v is an eigenvector with eigenvalue λ , then $\overline{v}$ is an eigenvector with eigenvalue $\overline{\lambda}$.

Definition 46.4

Let A be an $m \times n$ -matrix. The conjugate transpose A^H of A is defined as $\overline{A}^T$. We say the matrix A is Hermitian if $A = A^H$.